DA 54. Analysis of changes to the SPGZ in 2025 in terms of grounds

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Install packages and attach libraries

```
# install.packages("psych")
# install.packages("lubridate")
# install.packages("purrr")
# install.packages("GGally")
# install.packages("skimr")
# install.packages("Writexl")
# install.packages("corrplot")
library(psych)
library(googlesheets4)
library(tidyr)
```

```r
library(dplyr)
library(janitor)
library(ggplot2)
library(lubridate)
library(purrr)
library(GGally)
library(readxl)
library(skimr)
library(writexl)
library(corrplot)
```

1. Business task, context and scope of work

1.1. Key tasks and deliverable(s)

1.1.1. Business task

Determine the number of changes made to the SPGZ in 2025 for each of possible scenarios (cases) for the subsequent determination of frequency of scenarios (cases) in which customers receive notifications (in accordance with "EAIST_JB-8765 / , ") and scenarios (cases) that entail the need for customers to make changes to the lot specifications.

Notes

- The assignment to carry out of this task was issued at the "Meet-145 , ( )" meeting.
- This task is carrying out in the card "DA-54 / / 2025 "
- The results of this task will be used in the following tasks:
  - SGL-105 ( )
  - Meet-145 , ( )

1.1.2. Key stakeholders

The key stakeholders of this task are following:

- Department of Economic Policy and Development of the City of Moscow
- Department of Competition Policy of the City of Moscow
- Moscow city customers

1.1.3. SMART questions

The result of completing this task should be the answers to the following questions

N	Type	Question
1	- Specific - Measurable - Relevant - Time-bound	How often did the SPGZ change in 2025 for each of the scenarios (cases) described in " " file, prepared during the work on task "IS_Task-318 , ".
2	...	...

1.1.4. Scope of work (SOW)

N	Deliverables	Timeline	Milestones	Results	What SMART questions will be answered
1	Dataset with information about changes SPGZ in 2025 from Catalog database	18.12.2025	Completion of extracting data from Catalog database	Dataset with information about changes SPGZ in 2025 from Catalog database	-
2	Clean dataset	19.12.2025	Dataset is clean and usable	-	
3	Exploratory data analysis (EDA)	21.12.2025	Completion of EDA - understanding of dataset is formed	EDA highlights room for investigation. Questions were formulated based on the results of EDA.	-
4	Engineered features in the form of new columns	22.12.2025	New columns engineered to provide more accessible information		
5	Statistical insights	24.12.2025	Completion of statistical analysis	Statistical analysis conducted	How often did the SPGZ change in 2025 for each of the scenarios (cases) described in “ ” file, prepared during the work on task “IS_Task-318 , ”.
6	Visual aids	25.12.2025	Completion of visualizations	Visualizations created	-
7	Final report	27.12.2025	Final report submission	Final report completed	-

1.2 Guiding questions

N	Question	Answer
1	What problem need to be solved using the result of this task	The problem that needs to be solved is that customers need to make changes in they lots if the SPGZ positions used in those lot’s specifications are changed or deleted
2	How can the insights (results) of this task drive business decisions	This analysis insights will be used to determine necessary EAIST upgrades and their priorities.

2. Data preparation

2.1. Key tasks and deliverables

2.1.1. Collecting the necessary data

Data was collected from several Catalog database tables using the SQL query "spgz__kpgz__relations__packages.sql".

2.1.2. Downloading and storing data

The data was downloaded and stored at the enterprise cloud and available is at this link

```
# df <- read.csv("C:/Users/GudievZK/Nextcloud/Data analysis/DA-54/2025.12.18_spgz_kpgz_ktru_pack_rels_d
# df_src <- read.csv("C:/Users/GudievZK/Nextcloud/Data analysis/DA-54/2025.12.18_spgz_kpgz_ktru_pack_re


df <- read_excel("/Users/zelimkhan/Desktop/DA-54/2025.12.18_spgz_kpgz_ktru_pack_rels_db_ktlg.xlsx", gues
df_src <- read_excel("/Users/zelimkhan/Desktop/DA-54/2025.12.18_spgz_kpgz_ktru_pack_rels_db_ktlg.xlsx",
```

2.1.3. Data organization

The data is organized as a table (in csv format) containing **58682** rows. The rows contain information about versions of SPGZ positions which were changed or deleted in 2025.

2.1.4. Data credibility determination

The ROCCC framework and Data integrity methodology were used to determine the data credibility

ROCCC framework

N	ROCCC Dimension	Description	Conclusion	Comments
1	Reliable	Source is dependable and consistent.	See to the conclusion below the table	It depends on ROCCC dimensions listed below
2	Original	The data obtained from the original source (or if the data is not obtained from the original source, it is possible compere and verify the data with the original source)	The data obtained from the original source (Catalog database).	
3	Comprehensive	Covers all necessary cases and variables	The data covers all necessary cases and variables	
4	Current	The data source aren't out of date or irrelevant	The data is current and contains the necessary information for 2025 year	
5	Cited	The data source is cited and vetted	The data source is cited and veted	

Conclusion on reliability of the data source based on the on ROCCC framework: The data source is reliable.

Data integrity

N	Data integrity dimension	Description	Conclusion
1	Accuracy	The degree to which the data conforms to the actual entity being measured or described	The data conforms actual changes in the SPGZ in 2025
2	Completeness	Data completeness is a dimension of data quality that measures whether all necessary or required data is present and available for dataset to fulfill to intended purpose.	The dataset contains all necessary cases and variables.
3	Consistency	Data follows the same formats and rules.	The data is consistency
4	Trustworthiness	The trustworthiness refers to the degree to which information can be relied upon as accurate, credible and secure, allowing users confidently make decisions and conduct analysis based on it.	The data is trustworthiness

Conclusion on credibility of the data source based on Data integrity methodology: The source is credibility.

2.1.5. Bias

The dataset come from Catalog database and reflect actual changes in the SPGZ in 2025

2.1.6. Conclusion on the admissibility of using data

The data is credible and can be use for analysis

2.2 Guiding questions

N	Questions	Answers
1	Where the data is located?	The data was downloaded from Catalog database and stored in the enterprise cloud at this link (see section “2.1.2. Downloading and storing data”)
2	How is the data organized?	The data is organized as a table consisting of **58682** and **47**
3	Are there issues with bias or credibility in the data? Does the data ROCCC?	The data is unbiased and credible. Data is integrity and meets the ROCCC methodology requirements (see section the “2.1.4. Data credibility determination” section)
4	How are issues of licencing, privacy, security, and accessibility addressed?	

3. Data processing

Stage 1 of EDA: Data Discovery & Profiling

Table overview

```
df
```

```
## # A tibble: 58,682 x 47
##    TOTAL_ROWS ID_ENTITY_SPGZ_EAIST ID_SPGZ_EAIST ID_ENTITY_SPGZ_KATALOG
##    <chr>      <chr>                <chr>         <chr>
##  1 58682      10008020             1530772545    10008020
##  2 58682      10008020             1592957700    10008020
##  3 58682      10008215             1530772547    10008215
##  4 58682      10008215             1592956858    10008215
##  5 58682      10044442             10044442      10044442
##  6 58682      10044442             1640044123    10044442
##  7 58682      10052745             1530785272    10052745
##  8 58682      10052745             1575153004    10052745
##  9 58682      10052764             1530785269    10052764
## 10 58682      10052764             1575153001    10052764
## # i 58,672 more rows
## # i 43 more variables: ID_VERSION_SPGZ_KATALOG <chr>, PARENTID_SPGZ <chr>,
## #   CODE_KPGZ_SPGZ <lgl>, NAME_SPGZ <chr>, ISSTANDARDIZED_SPGZ <chr>,
## #   ISPROJECT_SPGZ <chr>, ISDELETED_SPGZ <chr>, ISEXCLUDED_SPGZ <chr>,
## #   ISEMIAS_SPGZ <chr>, SPGZATTRIBUTES_SPGZ <chr>, ADDITIONALUNIT_SPGZ <chr>,
## #   ISPCP_SPGZ <chr>, ATTRIBUTETARIF_SPGZ <chr>, KTRUID_SPGZ <chr>,
## #   OKPDID_SPGZ <chr>, ISKPGZ_SPGZ <chr>, CREATEDATE_SPGZ <chr>, ...
```

Clean names

```
# toString(names(df))
df <- clean_names(df)
```

Columns names

```
names(df)
```

```
##  [1] "total_rows"              "id_entity_spgz_eaist"
##  [3] "id_spgz_eaist"           "id_entity_spgz_katalog"
##  [5] "id_version_spgz_katalog" "parentid_spgz"
##  [7] "code_kpgz_spgz"          "name_spgz"
##  [9] "isstandardized_spgz"     "isproject_spgz"
## [11] "isdeleted_spgz"          "isexcluded_spgz"
## [13] "isemias_spgz"            "spgzattributes_spgz"
## [15] "additionalunit_spgz"     "ispcp_spgz"
## [17] "attributetarif_spgz"     "ktruid_spgz"
## [19] "okpdid_spgz"             "iskpgz_spgz"
## [21] "createdate_spgz"         "editdate_spgz"
## [23] "begindate_version_spgz"  "enddate_version_spgz"
## [25] "is_kpgz"                 "okpd2_id_spgz"
## [27] "okpd2_code_spgz"         "okpd2_name_spgz"
## [29] "okpd2_canceldate_spgz"   "okpd2_isactual_spgz"
## [31] "okpd2_okpd2id_spgz"      "ktru_code_spgz"
## [33] "ktru_name_spgz"          "rels_old_pgz_id"
## [35] "rels_linktype"           "rels_new_pgz_id"
## [37] "rels_is_actual"          "rels_previous_pgz_id"
```

```
## [39] "rels_source_pgz_id"      "package_name"
## [41] "pgzpack_pgz_id"          "package_reason_id"
## [43] "package_user_name"       "package_reason_name"
## [45] "id_entity_kpgz"          "name_kpgz"
## [47] "code_kpgz"
```

Number of rows and columns

```r
dim(df)
```

```
## [1] 58682    47
```

Data types

```r
glimpse(df)
```

```
## Rows: 58,682
## Columns: 47
## $ total_rows                <chr> "58682", "58682", "58682", "58682", "58682", "~
## $ id_entity_spgz_eaist      <chr> "10008020", "10008020", "10008215", "10008215"~
## $ id_spgz_eaist             <chr> "1530772545", "1592957700", "1530772547", "159~
## $ id_entity_spgz_katalog    <chr> "10008020", "10008020", "10008215", "10008215"~
## $ id_version_spgz_katalog   <chr> "30040471", "30384089", "30040473", "30384100"~
## $ parentid_spgz             <chr> "3206701", "3206701", "3206701", "3206701", "6~
## $ code_kpgz_spgz            <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ name_spgz                 <chr> "                         ~
## $ isstandardized_spgz       <chr> "1", "1", "1", "1", "0", "0", "0", "0", "0", "~
## $ isproject_spgz            <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ isdeleted_spgz            <chr> "0", "1", "0", "1", "0", "1", "0", "1", "0", "~
## $ isexcluded_spgz           <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ isemias_spgz              <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ spgzattributes_spgz       <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ additionalunit_spgz       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ ispcp_spgz                <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ attributetarif_spgz       <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ ktruid_spgz               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ okpdid_spgz               <chr> "39566", "39566", "39566", "39566", "44818", "~
## $ iskpgz_spgz               <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ createdate_spgz           <chr> "2024-08-26 18:10:59.000", "2025-03-19 15:41:3~
## $ editdate_spgz             <chr> "2024-08-26 16:08:59.092", "2025-03-19 15:07:3~
## $ begindate_version_spgz    <chr> "2024-08-26 00:00:00.000", "2025-03-19 00:00:0~
## $ enddate_version_spgz      <chr> "2025-03-19 00:00:00.000", NA, "2025-03-19 00:~
## $ is_kpgz                   <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ okpd2_id_spgz             <chr> "1750486", "2063125", "1750486", "2063125", "1~
## $ okpd2_code_spgz           <chr> "28.99.39.190", "28.99.39.190", "28.99.39.190"~
## $ okpd2_name_spgz           <chr> "                        , ~
## $ okpd2_canceldate_spgz     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ okpd2_isactual_spgz       <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "~
## $ okpd2_okpd2id_spgz        <chr> "8888313", "8888313", "8888313", "8888313", "8~
## $ ktru_code_spgz            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ ktru_name_spgz            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_old_pgz_id           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_linktype             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_new_pgz_id           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_is_actual            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_previous_pgz_id      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
```

```
## $ rels_source_pgz_id      <chr> "10050054", "10050054", "10050055", "10050055"~
## $ package_name            <chr> "   17", "                        ~
## $ pgzpack_pgz_id          <chr> "30040471", "30384089", "30040473", "30384100"~
## $ package_reason_id       <dbl> 5, 5, 5, 5, NA, 3, 5, 5, 5, 5, 5, 5, 5, 5, 5, ~
## $ package_user_name       <chr> "                  ", "           ~
## $ package_reason_name     <chr> "         ", "         ", "        ~
## $ id_entity_kpgz          <dbl> 3206701, 3206701, 3206701, 3206701, 68912, 689~
## $ name_kpgz               <chr> "                      ", "      ~
## $ code_kpgz               <chr> "01.25.93.12", "01.25.93.12", "01.25.93.12", "~
```

Data classes

```r
sapply(df, class)
```

```
##              total_rows     id_entity_spgz_eaist            id_spgz_eaist
##             "character"              "character"              "character"
##  id_entity_spgz_katalog id_version_spgz_katalog            parentid_spgz
##             "character"              "character"              "character"
##          code_kpgz_spgz                name_spgz      isstandardized_spgz
##               "logical"              "character"              "character"
##           isproject_spgz           isdeleted_spgz           isexcluded_spgz
##             "character"              "character"              "character"
##             isemias_spgz     spgzattributes_spgz      additionalunit_spgz
##             "character"              "character"              "character"
##              ispcp_spgz      attributetarif_spgz               ktruid_spgz
##             "character"              "character"              "character"
##             okpdid_spgz              iskpgz_spgz           createdate_spgz
##             "character"              "character"              "character"
##           editdate_spgz   begindate_version_spgz     enddate_version_spgz
##             "character"              "character"              "character"
##                 is_kpgz            okpd2_id_spgz           okpd2_code_spgz
##             "character"              "character"              "character"
##         okpd2_name_spgz    okpd2_canceldate_spgz      okpd2_isactual_spgz
##             "character"                "logical"              "character"
##      okpd2_okpd2id_spgz           ktru_code_spgz            ktru_name_spgz
##             "character"              "character"              "character"
##         rels_old_pgz_id            rels_linktype           rels_new_pgz_id
##             "character"                "numeric"              "character"
##          rels_is_actual    rels_previous_pgz_id        rels_source_pgz_id
##             "character"                "logical"              "character"
##             package_name           pgzpack_pgz_id        package_reason_id
##             "character"              "character"                "numeric"
##       package_user_name      package_reason_name           id_entity_kpgz
##             "character"              "character"                "numeric"
##               name_kpgz                code_kpgz
##             "character"              "character"
```

Checking duplicates. Number of duplicate columns in each column.

```r
sapply(df, function(x) sum(duplicated(x)))
```

```
##              total_rows     id_entity_spgz_eaist            id_spgz_eaist
##                   58681                    41066                    17368
##  id_entity_spgz_katalog id_version_spgz_katalog            parentid_spgz
##                   41114                    16702                    56644
##          code_kpgz_spgz                name_spgz      isstandardized_spgz
```

```
##                    58681                   45324                   58680
##          isproject_spgz           isdeleted_spgz          isexcluded_spgz
##                    58681                   58680                   58680
##             isemias_spgz     spgzattributes_spgz      additionalunit_spgz
##                    58681                   58680                   58680
##               ispcp_spgz     attributetarif_spgz              ktruid_spgz
##                    58679                   58681                   49485
##              okpdid_spgz             iskpgz_spgz          createdate_spgz
##                    58132                   58681                   37490
##            editdate_spgz  begindate_version_spgz    enddate_version_spgz
##                    49115                   57627                   55366
##                  is_kpgz           okpd2_id_spgz          okpd2_code_spgz
##                    58681                   54003                   56779
##          okpd2_name_spgz   okpd2_canceldate_spgz      okpd2_isactual_spgz
##                    56786                   58681                   58680
##        okpd2_okpd2id_spgz          ktru_code_spgz           ktru_name_spgz
##                    56779                   55178                   57733
##          rels_old_pgz_id           rels_linktype          rels_new_pgz_id
##                    45371                   58678                   57115
##           rels_is_actual    rels_previous_pgz_id       rels_source_pgz_id
##                    58679                   58681                   32038
##             package_name          pgzpack_pgz_id        package_reason_id
##                    54331                   17298                   58676
##        package_user_name     package_reason_name           id_entity_kpgz
##                    58618                   58676                   56644
##                name_kpgz               code_kpgz
##                    56644                   56644
```

Checking uniqueness.

Number of unique rows.

```r
df %>% n_distinct()
```

```
## [1] 58544
```

Number of unique values in each column

```r
sapply(df, function(x) sum(n_distinct(x)))
```

```
##               total_rows     id_entity_spgz_eaist            id_spgz_eaist
##                        1                    17616                    41314
##   id_entity_spgz_katalog  id_version_spgz_katalog            parentid_spgz
##                    17568                    41980                     2038
##           code_kpgz_spgz                name_spgz      isstandardized_spgz
##                        1                    13358                        2
##           isproject_spgz           isdeleted_spgz          isexcluded_spgz
##                        1                        2                        2
##             isemias_spgz     spgzattributes_spgz      additionalunit_spgz
##                        1                        2                        2
##               ispcp_spgz     attributetarif_spgz              ktruid_spgz
##                        3                        1                     9197
##              okpdid_spgz             iskpgz_spgz          createdate_spgz
##                      550                        1                    21192
##            editdate_spgz  begindate_version_spgz    enddate_version_spgz
##                     9567                     1055                     3316
##                  is_kpgz           okpd2_id_spgz          okpd2_code_spgz
```

```
##                        1                     4679                     1903
##          okpd2_name_spgz   okpd2_canceldate_spgz      okpd2_isactual_spgz
##                     1896                        1                        2
##       okpd2_okpd2id_spgz          ktru_code_spgz           ktru_name_spgz
##                     1903                     3504                      949
##          rels_old_pgz_id          rels_linktype          rels_new_pgz_id
##                    13311                        4                     1567
##           rels_is_actual    rels_previous_pgz_id      rels_source_pgz_id
##                        3                        1                    26644
##             package_name           pgzpack_pgz_id        package_reason_id
##                     4351                    41384                        6
##        package_user_name     package_reason_name           id_entity_kpgz
##                       64                        6                     2038
##                name_kpgz                code_kpgz
##                     2038                     2038
```

Checking completness

```r
colSums(is.na(df))
```

```
##               total_rows     id_entity_spgz_eaist             id_spgz_eaist
##                        0                       75                        61
##   id_entity_spgz_katalog id_version_spgz_katalog             parentid_spgz
##                        0                        0                         0
##           code_kpgz_spgz                name_spgz       isstandardized_spgz
##                    58682                        0                         0
##           isproject_spgz           isdeleted_spgz           isexcluded_spgz
##                        0                        0                         0
##             isemias_spgz     spgzattributes_spgz       additionalunit_spgz
##                        0                        0                     58681
##               ispcp_spgz       attributetarif_spgz              ktruid_spgz
##                     1633                        0                     42773
##              okpdid_spgz              iskpgz_spgz           createdate_spgz
##                    49819                        0                         0
##            editdate_spgz   begindate_version_spgz     enddate_version_spgz
##                     1779                        0                     19829
##                  is_kpgz            okpd2_id_spgz           okpd2_code_spgz
##                        0                        0                         0
##          okpd2_name_spgz   okpd2_canceldate_spgz       okpd2_isactual_spgz
##                        0                    58682                         0
##       okpd2_okpd2id_spgz          ktru_code_spgz            ktru_name_spgz
##                        0                    47586                     47587
##          rels_old_pgz_id           rels_linktype           rels_new_pgz_id
##                    40655                    40655                     40655
##           rels_is_actual    rels_previous_pgz_id        rels_source_pgz_id
##                    40655                    58682                     22851
##             package_name           pgzpack_pgz_id         package_reason_id
##                      613                      613                       613
##        package_user_name     package_reason_name            id_entity_kpgz
##                     3638                      613                         0
##                name_kpgz                code_kpgz
##                        0                        0
```

Converting data types Converting all columns with fewer than 10 unique (non-NA) values into factors

```r
# df <- df %>%
#   mutate(
#     across(
#       where(~ n_distinct(.x, na.rm = TRUE) < 10),
#       as.factor
#     )
#     )
```

Converting variables with date and time values from character format to date format.

```r
df <- df %>%
  mutate(
    across(
      contains("date_") & where(is.character),
           ~ as.Date(ymd_hms(.x)))
  )
```

Filtering data

```r
df <- df %>%
  filter(
    (isexcluded_spgz == 0 & !is.na(package_user_name)) & # deleting excluded spgz and spgz without auth
      (
          year(editdate_spgz) == 2025 |
        year(enddate_version_spgz) == 2025)
        )
```

Deleting duplicate rows

```r
df <- df %>%
   distinct()
```

Deleting unnecessary columns

```r
df <- df %>%
  select(id_entity_spgz_katalog, id_version_spgz_katalog, name_spgz, isstandardized_spgz, isproject_spgz
  ispcp_spgz, createdate_spgz, editdate_spgz, begindate_version_spgz, enddate_version_spgz, okpd2_code_s
  okpd2_name_spgz, ktru_code_spgz, ktru_name_spgz, rels_old_pgz_id, rels_linktype, rels_new_pgz_id, rels
  rels_source_pgz_id, package_name, package_user_name, package_reason_name, id_entity_kpgz, name_kpgz, c

# toString(names(df))
```

```r
dups <- df %>%
  group_by(across(everything())) %>%  # group by all columns
  filter(n() > 1) %>%                 # keep groups with more than one row
  ungroup()
```

Adding new columns Adding a column with number of values for each id_entity_spgz_katalog

```r
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    n_id_entity_spgz_katalog = n()) %>%
  ungroup()
```

Adding a column with the numbers of unique values id_version_spgz_katalog per id_entity_spgz_katalog

```r
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    n_unique_id_version_cataloge = n_distinct(id_version_spgz_katalog, na.rm = TRUE)) %>%
      ungroup()
```

Adding column with difference between "id_entity_spgz_katalog" and "id_version_spgz_katalog"

```r
df <- df %>%
  mutate(
    diff_id_entity_id_version = n_id_entity_spgz_katalog - n_unique_id_version_cataloge
  )
```

Adding a column with deletion information

```r
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    is_deleted_spgz = as.integer(sum(as.integer(isdeleted_spgz), na.rm = TRUE) > 1)
    ) %>%
      ungroup()
```

Adding column with information about relation…

Adding column with information about standardizing in 2025

```r
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    was_standartized_25 = as.integer(
      n_distinct(trimws(as.character(isstandardized_spgz)), na.rm = TRUE) > 1
      )
    ) %>%
  ungroup()
```

dssd

```r
# rels_old_pgz_id <- df %>% filter(!is.na(rels_old_pgz_id)) %>% select(rels_old_pgz_id)
# df_rels <- df %>% filter(id_version_spgz_katalog %in% rels_old_pgz_id)

# df_0 %>% filter(
#   id_version_spgz_katalog == "30425543" |
#     rels_old_pgz_id == "30425543" |
#     id_entity_spgz_katalog == "29270927" |
#     id_entity_spgz_katalog == "30424361" |
#     id_version_spgz_katalog == "30425542"
# ) %>% # write_xlsx("/Users/zelimkhan/Desktop/DA-54/2025.12.21_df_0.xlsx")
#    # write_xlsx("C:/Users/GudievZK/Nextcloud/Data analysis/DA-54/df_0.xlsx")
```

```
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# **4. Data analysis**
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```

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# **5. Report**
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# **6. Usage result**


``` r
# write_xlsx(df, "/Users/zelimkhan/Desktop/DA-54/2025.12.21_df.xlsx")
~~~~